



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CS3501 Compiler Design

Name of the Faculty member (s): Dr.I.Gethzi Ahila Poornima , AP(SG)/CSE

Innovative Practice Description

- **Unit / Topic: Unit III / Translation of Expressions and Type Checking, Back Patching**
- **Course Outcome: CO3**
- **Activity Chosen: One Minute paper**

- **Justification:**

In order to produce optimized machine dependant code, the compiler designer should design the compiler so that it could translate the expression in the given source code effectively and ensure that the type checking done effectively based on the language. Without the knowledge of translating expression, type checking and back patching, the further procedures in designing compiler cannot be performed effectively.

- **Time Allotted for the Activity: 10 Minutes**
- **Details of the Implementation:**

Steps to translate expressions and concept of type checking and back patching are explained clearly in the first 40 minutes. A sample code segment with expressions was considered and the way to translate the expression and the concept of type checking and backtracking was explained in detail in white board and using powerpoint presentation in projector. At the end of the session, all the students were asked to write down the concept of translating expressions, type checking and backpatching. They were asked not to copy others and told them that this was not a typical assessment which includes mark. As a teacher, I made them to understand that, this write up is for the faculty to understand the level of understanding the concept since this is one of the most important concept a student should learn. The understanding level of students were determined by going through the paper written by them. Figure 1 shows paper written by Ms SugirthaSri (Reg.No: 953621104048) . Exactly 5 of the students had trouble in translating the expression itself.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PSO1
CO3	3	3	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PSO1
	3	1	1	1

Justification for correlation	The idea of translating expression and type checking will make the students to design a compiler.	The students examine the translation of expression and backpatching being done during the translation process.	The students find the solutions for designing compiler especially semantic analyzer phase.	The Student's concise writing of their questions helped them develop their skills.	The students apply expression translation and type checking in developing a compiler.
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• **Images / Screenshot of the practice:**

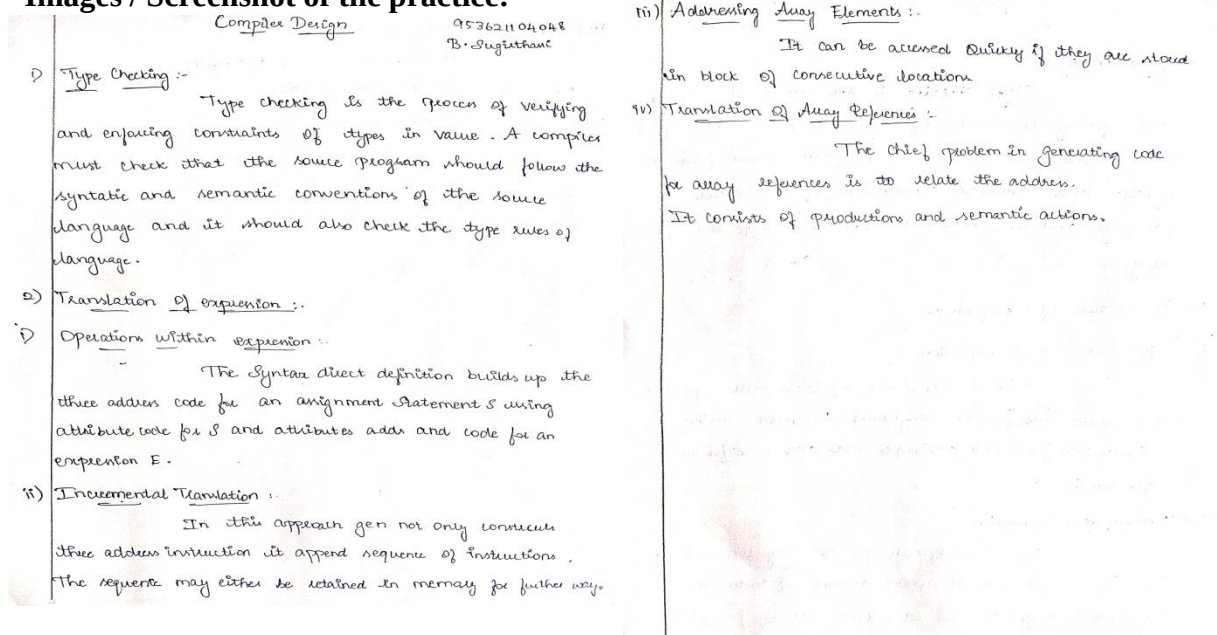


Figure 1: Sample sheet written by Ms. Sugirtha Sri -953621104048

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

The students gave this activity positive reviews. They said they were able to recall the day's lesson and describe its difficulty.

❖ **Benefit of the practice:**

- Most of the students take this activity as positive and started solving the conversion process.
- I was capable of estimating the students' degree of comprehension in my capacity as faculty.

❖ **Challenges faced in implementation:**

- Very few students hesitated to recollect the steps involved in the conversion.

❖ **References:**

- <https://www.slideshare.net/raja2dlas/minute-paper>
- <https://www.ritrjpm.ac.in/images/computer-science/CS8251-SE-MinutePaper.>

Signature of Faculty Member

HOD